User Guide for OpenFOAM-8

porousPBRFoam: An OpenFOAM-based solver for porous-media modeling of catalytic packed-bed reactors with detailed surface chemistry

Please cite our article if you are using this package: J. H. Lee, D. N. Nguyen, H. W. Seo, H. J. Ahn, C. S. Yoo, porousPBRFoam: An OpenFOAM-based solver for porous-media modeling of catalytic packed-bed reactors with detailed surface chemistry (2026) (submitted).

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1. General descriptions

The framework is fully native OpenFOAM(OF) and comprises newly developed libraries and solvers as follows:

- libraries:
  - **thermophysicalModels** includes surface chemistry sub-library.
- solvers:
  - **porousPBRFoam**
- utilities:
  - **surfChemkinToFoam**

1.1. Library for surface chemistry in OpenFOAM

Figure 1 presents the class diagram of the updated `thermophysicalModels` library, including the `surfaceChemistryModels` sub-library used for heterogeneous surface chemistry. The thermophysical library developed in this work extends the `surfaceChemistryFoam` framework [1] to support porous-media simulations of catalytic packed-bed reactors. The resulting class structure is presented in Fig. 1. Detailed descriptions of the classes inherited from `surfaceChemistryFoam` are provided in *surfaceChemistryFoam_ userGuide_ OF8.pdf*, available in the `porousPBRFoam-8/documentations` directory [2]. The present section focuses on the classes introduced for PMM.

- *PorousMediaSurfaceChemistryModel* incorporates detailed heterogeneous surface chemistry into the porous-media formulation. It solves the coupled cell-local ODE systems for the gas-phase and surface species in each porous cell. The resulting surface-reaction rates are scaled by the catalytic surface-to-volume ratio, F_{S2V}, to obtain the volumetric species-production and heat-release source terms, $\dot{R}_i^{\mathrm{cat}}$ and $\dot{Q}^{\mathrm{cat}}$, incorporated into the macroscopic species and energy conservation equations, respectively.

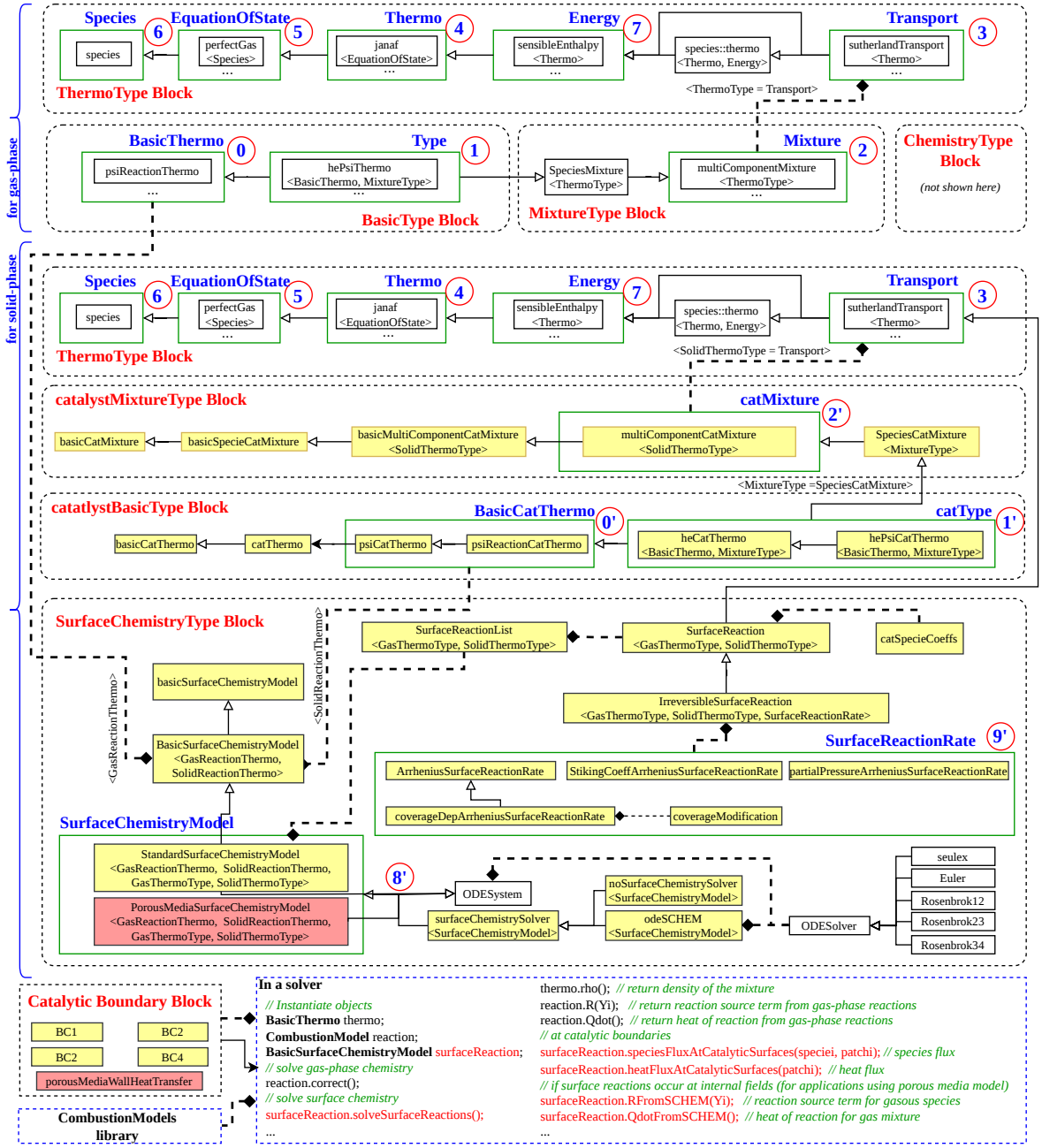


Figure 1: Class diagram of the extended thermophysical library. Yellow boxes denote classes inherited from `surfaceChemistryFoam`, whereas red boxes denote classes introduced in the present work for PMM. Boxes with circled numbers indicate representative classes selected through the run-time selection mechanism. Solid and dashed arrows indicate inheritance and usage relationships, respectively.

- *porousMediaWallHeatTransfer* provides a wall boundary condition for heat transfer between the reactor wall and the packed bed. It evaluates the wall heat flux using the wall-to-bed heat-transfer coefficient, h_w.

1.2. Solver and preprocessing utility

`porousPBRFoam` solver provides functionality for non-reacting and reacting porous-media simulations of packed-bed reactors (PBRs). It solves the macroscopic mass, momentum, energy, and species conservation equations. Detailed heterogeneous surface chemistry can be enabled for catalytic reacting-flow simulations.

`surfChemkinToFoam` utility is an extension of the standard OpenFOAM `chemkinToFoam` utility to provide functionality for converting heterogeneous surface-chemistry input files in CHEMKIN format (`surf.inp` and `surfTherm.dat`) into OpenFOAM dictionaries (`reactions.surf` and `thermo.solid`).

2. Case-file description for non-reacting flow simulation

The first tutorial, a wall-heated packed-bed case, demonstrates non-reacting flow and heat-transfer simulation in a packed-bed reactor (PBR) using `porousPBRFoam` with a constant-density model (`rhoConst`). The case is available in `tutorials/01_WallHeated_HeatTransfer` within the `porousPBRFoam-8` source repository [2] and corresponds to the wall-heated heat-transfer validation case presented in the manuscript. Its configuration follows the standard OpenFOAM directory structure, as shown in Fig. 2. An additional `porousMediaProperties` dictionary is required to specify the packed-bed properties and porous-media closure models.

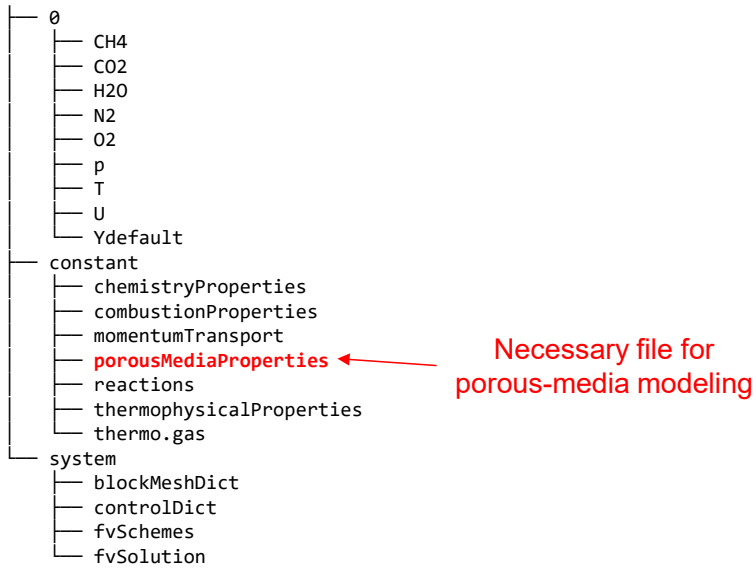


Figure 2: Case file structure for the non-reacting porous media heat-transfer simulation.

2.1. Settings at time directory

The initial fields are configured as in a standard `reactingFoam` case, except that the temperature field requires the `porousMediaWallHeatTransfer` boundary condition to model wall-to-bed heat transfer.

Setting for the temperature field. The wall boundary condition for the `T` should be set to `porousMediaWallHeatTransfer`, as shown below, when convective heat transfer between the reactor wall and the packed bed is considered. For an adiabatic wall, the `zeroGradient` boundary condition can be used.

```
/*---------------------------*- C++ -*-----------------------------*\
  =========                 |
  \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox
   \\    /   O peration     | Website:  https://openfoam.org
    \\  /    A nd           | Version:  8
     \\/     M anipulation  |
\*-------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       volScalarField;
    location    "0";
    object      T;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //
```

```
dimensions      [0 0 0 1 0 0 0];

internalField   uniform 293;

boundaryField
{
    inlet
    {
        type            fixedValue;
        value           uniform 293;
    }
    wall
    {
        type            porousMediaWallHeatTransfer;
        mode            coefficient;
        kappaMethod     fluidThermo;
        h               uniform 100;
        Ta              constant 373;
        value           $internalField;
    }
    ...
}
// ************************************************************************* //
```

2.2. Settings at constant directory

Setting for porousMediaProperties file. The `porousMediaProperties` dictionary specifies solid properties, bed geometry, and closure models for momentum resistance ($\mathbf{S}_m$), thermal conductivity (k_{eff}), species diffusivity ($D_{i,\text{eff}}$), and wall-to-bed heat transfer (h_w), along with model-specific parameters. An example for the non-reacting heat-transfer case is shown below.

```
/*--------------------------*- C++ -*----------------------------------*\
  =========                 |
  \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox
   \\    /   O peration     | Website:  https://openfoam.org
    \\  /    A nd           | Version:  8
     \\/     M anipulation  |
\*---------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       dictionary;
    location    "constant";
    object      porousMediaProperties;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //
// 1. Simulation controls
solverMode                                          PM;
isothermal                                          off;

// 2. Solid properties and bed geometry
```

```
SolidThermalConductivity                    1.175;    // [W/m-K]
SolidDensity                                2719;    // [kg/m3]
SolidHeatCapacity                           871;     // [J/kg-K]
Porosity                                    0.4165; // [-]
ParticleDiameter                            5e-3;    // [m]

// 3. Porous-media closure model
pressureDropClosureModel                    None;
ThermalConductivityClosureModel             Const;
DiffusivityClosureModel                     BruggemanWilhelm;
WallHeatTransferClosureModel                Const;
EffectiveThermalConductivity                0.7;
heatTransferCoefficient                     100;

// ************************************************************************* //
```

The principal entries in the `porousMediaProperties` dictionary, including the solid-phase thermophysical properties and packed-bed geometrical and flow parameters, are described below.

- `solverMode`: `PMSCHEM` enables detailed heterogeneous surface chemistry, whereas `PM` disables it.
- `isothermal`: `off` (default) solves the macroscopic energy equation, whereas `on` retains the prescribed temperature field without solving the energy equation.
- `SolidThermalConductivity` (k_s) specifies the thermal conductivity of the solid particle.
- `SolidDensity` (ρ_s) specifies the density of the solid particle.
- `SolidHeatCapacity` ($c_{p,s}$) specifies the specific heat capacity of the solid particle.
- `Porosity` (ε) specifies the porosity (void fraction) of the packed bed.
- `ParticleDiameter` (d_p) specifies the diameter of solid particle.

The porous-media closure models and their roles are summarized below. Model-specific input parameters are summarized in Section 4.

- `pressureDropClosureModel` selects the momentum-resistance closure model. Available options are *None*, *Darcy* [3], *Forchheimer* [4], and *Ergun* [5]. The *None* option disables momentum resistance.

- `DiffusivityClosureModel` supports the *BruggemanWilhelm* model [6–8] for evaluating the effective species diffusivity ($D_{i,\text{eff}}$).

- `ThermalConductivityClosureModel` selects the model for evaluating the effective thermal conductivity (k_{eff}). Available options are *Const*, *ConstAnisotropic*, *Standard*, *Bunnell* [9], *Yagi* [10], and *Wakao* [11]. The *Wakao* model additionally requires a stagnant thermal conductivity model, specified through `StagnantThermalConductivityClosureModel`. Available options are *Kunii* [12], *Krupiczka* [13], and *ZS* [14].

- `WallHeatTransferClosureModel` selects the model for evaluating the wall-to-bed heat-transfer coefficient (h_w). Available options are *Const*, *Yagi* [15], *Li* [16], and *Bey* [17].

Setting for thermophysicalProperties file. The `thermophysicalProperties` dictionary defines the gas-phase thermophysical-model combination through `thermoType`. To use the `rhoConst` equation of state, specify the *Constant-density* configuration listed in Table 1. To use the `perfectGas` equation of state, specify the *Perfect-gas* configuration in the same table.

Table 1: ThermoType combinations for the `const` and `perfectGas` configurations.

Configuration	type	mixture	transport	thermo	energy	equationOfState	specie
Constant-density	heRhoThermo	multi	const	hConst	senE	rhoConst	specie
Perfect-Gas	hePsiThermo	multi	STM	janaf	senE/senI	perfectGas	specie

Abbreviations: multi: `multiComponentMixture`; STM: `sutherland`;
senE: `sensibleEnthalpy`; senI: `sensibleInternalEnergy`.

Setting the thermo.gas file. For the *Constant-density* `thermoType` configuration using `rhoConst`, specify the species properties following the O_2 example below. For the *Perfect-gas* configuration using `perfectGas`, refer to the reacting-flow case described in Section 3.

```
O2
{
    specie
    {
        molWeight       31.9988;
```

```
    }
    equationOfState
    {
        rho             0.946;
    }
    thermodynamics
    {
        Cp              1008;
        Hf              0;
    }
    transport
    {
        mu              2.174e-05;
        Pr              0.71;
    }
}
```

2.3. Settings at system directory

The files in the `system` directory specify the numerical schemes and simulation controls. They are configured in the same manner as in the standard OpenFOAM `rhoPimpleFoam` solver.

2.4. Running the non-reacting flow simulations

Once all preparations are complete, go to the non-reacting tutorial directory, generate the mesh, and run the solver:

```
$ cd ~/OpenFOAM/yourDirectory/porousPBRFoam-8
$ cd tutorials/01_WallHeated_HeatTransfer
$ blockMesh
$ porousPBRFoam
```

Alternatively, run `./Allrun` from the same directory.

3. Case-file description for reacting-flow simulation

The second tutorial, an isothermal NH_3 decomposition case, demonstrates reacting flow in a packed-bed reactor (PBR) using `porousPBRFoam` with detailed heterogeneous surface chemistry. The case is available in `tutorials/02_Isothermal_NH3Decomposition` within the `porousPBRFoam-8` source repository [2] and corresponds to the isothermal NH_3

decomposition validation case presented in the manuscript. The case follows the standard OpenFOAM directory structure, as shown in Fig. 3, but requires additional dictionaries and input/output files for detailed surface chemistry compared to the non-reacting case.

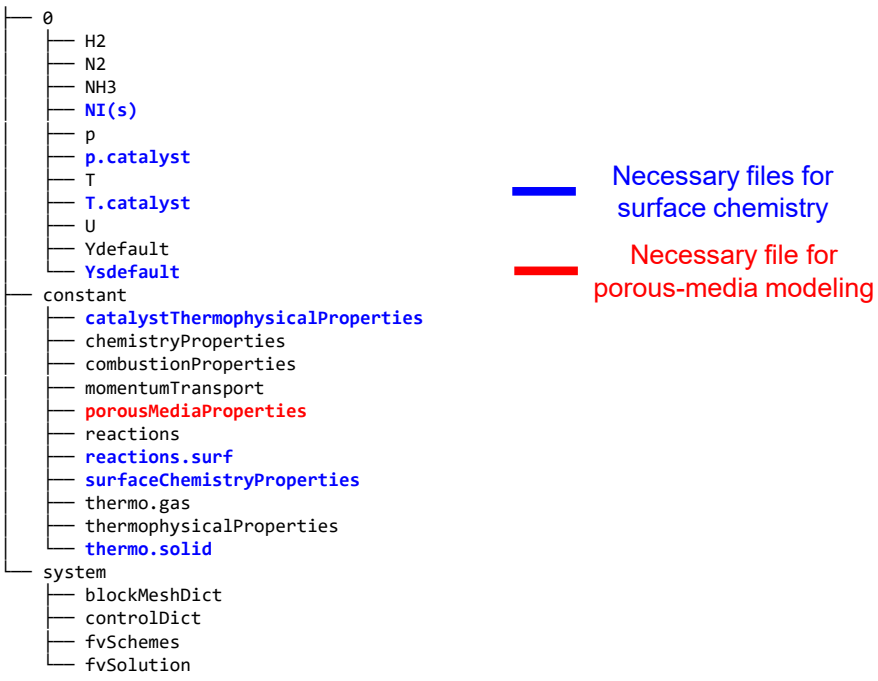


Figure 3: Case-file structure for a reacting-flow simulation using `porousPBRFoam` with heterogeneous surface chemistry. Additional files required for porous-media modeling and heterogeneous surface chemistry are highlighted separately.

In the initial-time directory (e.g., `0`), gas-phase fields are configured as in a standard `reactingFoam` case. Additional files are:

- `p.catalyst`: catalyst pressure.
- `T.catalyst`: catalyst temperature.
- `Ysdefault`: default initial site fractions.
- Individual site-species files (e.g., `NI(s)` and `Ru(s)`): initial fractions of the corresponding surface species.

The required site species depend on the reaction mechanism; intermediate site fractions are initialized to zero unless otherwise specified.

The additional files in `constant` are:

- `porousMediaProperties`: packed-bed properties, closure models, and controls for enabling detailed surface chemistry and selecting isothermal operation.
- `catalystThermophysicalProperties`: thermophysical-model configuration for surface (site) species.
- `surfaceChemistryProperties`: surface-chemistry calculation settings.
- `reactions.surf`: heterogeneous surface-reaction mechanism in OpenFOAM format.
- `thermo.solid`: thermodynamic data for surface (site) species in OpenFOAM format.

3.1. Settings at time directory

Surface-species site fractions. Initialize the site fractions to sum to unity, typically by setting the vacant-site species (e.g., `Ni(s)`) to 1.0 and adsorbed species to zero. Other initial distributions may be used, but setting all fractions to zero may cause initialization to fail.

```
/*---------------------------*- C++ -*-----------------------------*\
  =========                 |
  \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox
   \\    /   O peration     | Website:  https://openfoam.org
    \\  /    A nd           | Version:  8
     \\/     M anipulation  |
\*-------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       volScalarField;
    location    "0";
    object      Ni(s);
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //
dimensions      [0 0 0 0 0 0 0];

internalField   uniform 1;

boundaryField
{
    heatedWall
    {
```

```
        type            zeroGradient;
    }
    inlet
    {
        type            zeroGradient;
    }
    ...
}
// ************************************************************************* //
```

Settings for the temperature fields (T and T.catalyst). The gas-phase temperature field `T` is configured as described in Section 2.1. For detailed surface-chemistry simulations, an additional field, `T.catalyst`, must be provided. It can be created by copying `T`. Its values are not used in the PMM simulation.

Settings for the pressure fields (p and p.catalyst). The gas-phase pressure field `p` is configured as in a standard `reactingFoam` case. For detailed surface-chemistry simulations, an additional field, `p.catalyst`, must be provided. It can be created by copying `p` and setting its boundary conditions to `zeroGradient`. Its values are not used in the PMM simulation.

3.2. Settings at constant directory

Setting for porousMediaProperties file. The `porousMediaProperties` dictionary described in Section 2.2 requires several modifications for this reacting-flow case. In particular, `solverMode` must be set to `PMSCHEM` to enable detailed heterogeneous surface chemistry.

```
/*---------------------------*- C++ -*-----------------------------*\
  =========                 |
  \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox
   \\    /   O peration     | Website:  https://openfoam.org
    \\  /    A nd           | Version:  8
     \\/     M anipulation  |
\*-------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       dictionary;
    location    "constant";
    object      porousMediaProperties;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //
// 1. Simulation controls
solverMode                                          PMSCHEM;
isothermal                                          on;

// 2. Solid properties and bed geometry
```

```
SolidThermalConductivity                                30;      // [W/m-K]
SolidDensity                                            3540;    // [kg/m3]
SolidHeatCapacity                                       880;     // [J/kg-K]
Porosity                                                0.38;     // [-]
ParticleDiameter                                        2E-5;    // [m]

// Porous-media closure model
pressureDropClosureModel                                Forchheimer;
DiffusivityClosureModel                                 BruggemanWilhelm;
DarcyCoeff                                              1.0508E+07;
ForchCoeff                                              1.9773E+03;
// ************************************************************************* //
```

Setting for thermophysicalProperties file. The `thermophysicalProperties` dictionary defines the gas-phase thermophysical models following the standard `reactingFoam` format, as illustrated in Fig. 4(a). This tutorial uses the model combination listed in Table 2, with gas-mixture properties evaluated using the perfect-gas equation of state, JANAF thermodynamics, and Sutherland transport.

a) *thermophysicalProperties* file (for gas-phase)

```
thermoType      keyword      Selected model
{
  1 type            hePsiThermo;
  2 mixture         multiComponentMixture;
  3 transport       sutherland;
  4 thermo          janaf;
  5 energy          sensibleEnthalpy;
  6 equationOfState perfectGas;
  7 specie          specie;
}

inertSpecie AR;
#include "thermo.catalyst.gas"
```

b) *catalystThermophysicalProperties* file (for surface-phase)

```
solidThermoType   keyword      Selected model
{
  1' solidType          hePsiCatThermo;
  2' solidMixture       multiComponentCatMixture;
  3' solidTransport     const;
  4' solidThermo        hConst;
  5' solidEnergy        sensibleEnthalpy;
  6' solidEquationOfState rhoConst;
  7' solidSpecie        specie;
}

inertSpecie AR;
#include "thermo.catalyst.solid"
```

Figure 4: Example thermophysical settings for detailed surface-chemistry simulations using `porousPBRFoam`: (a) gas-phase models in `thermophysicalProperties` and (b) surface-phase models in `catalystThermophysicalProperties`.

Table 2: ThermoType combination for the heterogenous reacting flow simaultion tutorial.

type	mixture	transport	thermo	energy	equationOfState	specie
hePsiThermo	multi[a2]	STM[a3]	janaf	senE/senI[5]	perfectGas	specie

where all abbreviations are:

senE/senI[5]:`sensibleEnthalpy`/`sensibleInternalEnergy`;

multi[a2]: `multiComponentMixture`; STM[a3]: `sutherland`;

Setting for the catalystThermophysicalProperties file. The `catalystThermophysicalProperties` dictionary defines the thermophysical model combination for the surface species. The model combination is specified in `solidThermoType`, as illustrated in Fig. 4(b) and the example below. The corresponding species properties are read through `#include "thermo.solid"` from the `thermo.solid` file generated by `surfChemkinToFoam` utiliy, as described in Section 3.2.1.

```
/*---------------------------*- C++ -*-----------------------------*\
  =========                 |
  \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox
   \\    /   O peration     | Website:  https://openfoam.org
    \\  /    A nd           | Version:  8
     \\/     M anipulation  |
\*-------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       dictionary;
    location    "constant";
    object      catalystThermophysicalProperties;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

solidThermoType
{
    solidType            hePsiCatThermo;
    solidMixture         multiComponentCatMixture;
    solidTransport       const;
    solidThermo          hConst;
    solidEnergy          sensibleEnthalpy;
    solidEquationOfState rhoConst;
    solidSpecie          specie;
}

inertSpecie N2;

#include "thermo.solid"
// ************************************************************************* //
```

Setting for chemistryProperties file. The `chemistryProperties` dictionary is configured identically to that used in a standard `reactingFoam` case.

Setting for surfaceChemistryProperties file. The `surfaceChemistryProperties` dictionary specifies the model and ODE integration settings used to solve the coupled gas-phase and surface-species reaction system. It also defines the surface-to-volume ratio, F_{S2V}, used to scale surface-area-based reaction rates into volumetric species-production and heat-release

source terms. The reaction mechanism is read through `#include "reactions.surf"` from the `reactions.surf` file generated by `surfChemkinToFoam` utility, as described in Section 3.2.1.

```
/*---------------------------*- C++ -*-----------------------------*\
  =========                 |
  \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox
   \\    /   O peration     | Website:  https://openfoam.org
    \\  /    A nd           | Version:  8
     \\/     M anipulation  |
\*-------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       dictionary;
    location    "constant";
    object      surfaceChemistryProperties;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

// Select the porous-media surface-chemistry treatment
onlySurfaceCatalyst false; // default: false

// Compatibility entries; leave empty for the present PMM setup
catalystRegions
{
    catalyticSurfaces   ( );
    catalyticZones      ( );
}

// Catalytic surface area per unit reactor volume [1/m]
SurfaceToVolumeRatio            2.8E5;

// Surface-chemistry method and ODE solver
surfaceChemistryType
{
    method          porousMediaSCHEM;   // porous-media model
    solver          odeSCHEM;
}

chemistry           on;

initialChemicalTimeStep 1e-07;

odeSCHEMCoeffs
{
    solver          seulex;
    absTol          1e-8;
    relTol          0.01;
}

#include "reactions.surf"
// ************************************************************************* //
```

The main entries in `surfaceChemistryProperties` are:

- `onlySurfaceCatalyst`: set to `false` for Porous-media modeling.
- `catalystRegions`: set `catalyticSurfaces` and `catalyticZones` as empty.
- `SurfaceToVolumeRatio`: catalytically active surface area per unit volume of the packed bed, which is used to convert surface species-production and heat-release rates into volumetric source terms.
- `surfaceChemistryType`: set `method` to *porousMediaSCHEM* and `solver` to *odeSCHEM*.
- `odeSCHEMCoeffs`: ODE integrator and tolerances; the example uses `seulex` with absolute and relative tolerances of 10^{-8} and 10^{-2}.
- `chemistry`: enables or disables surface chemistry.
- `initialChemicalTimeStep`: initial time step for chemistry ODE integration.

3.2.1. Preparation of surface-chemistry input files

The surface-reaction mechanism and surface-species thermodynamic data must be generated in OpenFOAM format as `reactions.surf` and `thermo.solid`, respectively. From the `chemkin` directory, the `surfChemkinToFoam` utility converts the CHEMKIN input files and writes the resulting dictionaries to `constant`:

```
surfChemkinToFoam \
    surf.inp \
    surfTherm.dat \
    surfaceThermoDict \
    ../constant/reactions.surf \
    ../constant/thermo.solid
```

The required input files and an `Allchemkin` script that executes this command are provided in the tutorial. To run the script:

```
$ cd ~/OpenFOAM/yourDirectory/porousPBRFoam-8
$ cd tutorials/02_Isothermal_NH3Decomposition/chemkin
$ ./Allchemkin
```

The generated `reactions.surf` and `thermo.solid` dictionaries are included in `surfaceChemistryProperties` and `catalystThermophysicalProperties`, respectively. Alternatively, running `./Allrun` from the case directory performs input-file conversion, mesh generation, and solver execution in sequence.

Additional examples of OF format surface-chemistry mechanism file generation are provided in `tutorials/00_surfChemkinToFoam` within the source repository:

```
$ cd ~/OpenFOAM/yourDirectory/porousPBRFoam-8
$ cd tutorials/00_surfChemkinToFoam
```

```
1996_Deutschmann_Pt_CH4Oxidation/
2011_Appari_Ni_NH3Decomposition/
2011_Maier_Ni_SMR/
2013_Karakaya_Rh_H2Oxidation/
2013_Karakaya_Rh_H2Oxidation_example/
2014_Karakaya_Rh_WGS/
2016_Takahashi_Ru_NH3Decomposition_AKC/
2016_Takahashi_Ru_NH3Decomposition_PKC/
2021_Schmider_Ni_CO_CO2Methanation/
```

3.3. Settings at system directory

The files located in the `system` directory contain the numerical and simulation-control settings required for running the case. In general, their configuration is identical to that used in `reactingFoam`.

3.4. Running the reacting flow simulations

Navigate to the reacting-flow tutorial directory, generate the mesh, and run the solver:

```
$ cd ~/OpenFOAM/yourDirectory/porousPBRFoam-8
$ cd tutorials/02_Isothermal_NH3Decomposition
$ blockMesh
$ porousPBRFoam
```

Alternatively, run `./Allrun` from the same directory. These tutorials provide a starting point for configuring and running other PBR simulations.

4. Input parameters for porous-media closure models

This section summarizes the input parameters used to configure the porous-media closure models. Table 3 lists their keywords and symbols, distinguishing common inputs from model-specific inputs. Specify these entries in `constant/porousMediaProperties`, using the keyword names exactly as shown, including uppercase and lowercase letters. Table 4 lists the additional input parameters required by each closure model, excluding the common inputs listed in Table 3.

The momentum-exchange source, $\mathbf{S}_m$, is evaluated using the calculated superficial-velocity field. The Darcy and Forchheimer models require the prescribed resistance coefficients $\mathcal{D}$ and $\mathcal{F}$, as listed in Table 4, whereas the Ergun model calculates the resistance from ε and d_p together with the fluid properties.

The effective species diffusivity, $D_{i,\mathrm{eff}}$, is evaluated using the calculated superficial-velocity field. The flow-dependent correlations for k_{eff} and h_w instead use the reference superficial-velocity magnitude specified through `SuperficialVelocity`, denoted by $|\langle\mathbf{u}\rangle|$ in Table 4.

The inputs for evaluating k_{eff} are required when the energy equation is enabled. For *ConstAnisotropic*, the axial direction vector $\mathbf{e}_z$ must be nonzero and is normalized internally. For *Wakao*, `StagnantThermalConductivityClosureModel` must also be specified as *Kunii*, *Krupiczka*, or *ZS*.

The inputs for evaluating h_w apply when the `porousMediaWallHeatTransfer` boundary condition is used in `coefficient` mode. All wall heat-transfer models require k_s in the current implementation. The *Bey* model should be combined with *Wakao* to provide the stagnant thermal conductivity used in its calculation.

References

[1] D. N. Nguyen, J. H. Lee, H. W. Seo, H. J. Ahn, C. S. Yoo, surfaceChemistryFoam: An OpenFOAM-based library for detailed surface chemistry in reacting flow simulations, Comput. Phys. Commun. 327

Table 3: Input keywords and corresponding symbols for porous-media properties and closure models.

Symbol	Keyword	Unit
Common input parameters		
ε	`Porosity`	$[-]$
d_p	`ParticleDiameter`	$[\mathrm{m}]$
ρ_s	`SolidDensity`	$[\mathrm{kg\,m^{-3}}]$
$c_{p,s}$	`SolidHeatCapacity`	$[\mathrm{J\,kg^{-1}\,K^{-1}}]$
Model-specific input parameters		
$\mathcal{D}$	`DarcyCoeff`	$[\mathrm{m^{-2}}]$
$\mathcal{F}$	`ForchCoeff`	$[\mathrm{m^{-1}}]$
k_s	`SolidThermalConductivity`	$[\mathrm{W\,m^{-1}\,K^{-1}}]$
$\lvert\langle\mathbf{u}\rangle\rvert$	`SuperficialVelocity`	$[\mathrm{m\,s^{-1}}]$
k_{eff}	`EffectiveThermalConductivity`	$[\mathrm{W\,m^{-1}\,K^{-1}}]$
$k_{\mathrm{eff},r}$	`EffectiveRadialThermalConductivity`	$[\mathrm{W\,m^{-1}\,K^{-1}}]$
$k_{\mathrm{eff},z}$	`EffectiveAxialThermalConductivity`	$[\mathrm{W\,m^{-1}\,K^{-1}}]$
$\mathbf{e}_z$	`AxialDirection`	$[-]$
$(\alpha\beta)_{\mathrm{Y}}$	`YagiAlphaBeta`	$[-]$
φ_{Y}	`YagiPhi`	$[-]$
φ_{K}	`KuniiPhi`	$[-]$
γ_{K}	`KuniiGamma`	$[-]$
C_{mix}	`WakaoMixCoeff`	$[-]$
h_w	`heatTransferCoefficient`	$[\mathrm{W\,m^{-2}\,K^{-1}}]$
d_t	`ReactorDiameter`	$[\mathrm{m}]$

$\mathcal{D}$ and $\mathcal{F}$ denote the scalar resistance coefficients corresponding to $\boldsymbol{\mathcal{D}} = \mathcal{D}\mathbf{I}$ and $\boldsymbol{\mathcal{F}} = \mathcal{F}\mathbf{I}$ for isotropic resistance. The entry `SuperficialVelocity` specifies the reference superficial-velocity magnitude used in the selected closure models.

Table 4: Additional input parameters required by each porous-media closure model. A dash indicates that no additional input is required.

Closure quantity	Model	Additional input parameters
Momentum-exchange source, $\mathbf{S}_m$	*None*	–
	Darcy	$\mathcal{D}$
	Forchheimer	$\mathcal{D}$, $\mathcal{F}$
	Ergun	–
Effective species diffusivity, $D_{i,\mathrm{eff}}$	*BruggemanWilhelm*	–
Effective thermal conductivity, k_{eff}	*Const*	k_{eff}
	ConstAnisotropic	$k_{\mathrm{eff},r}$, $k_{\mathrm{eff},z}$, $\mathbf{e}_z$
	Standard	k_s
	Bunnell	$\lvert\langle\mathbf{u}\rangle\rvert$
	Yagi	k_s, $\lvert\langle\mathbf{u}\rangle\rvert$, $(\alpha\beta)_{\mathrm{Y}}$, φ_{Y}
	Wakao	k_s, $\lvert\langle\mathbf{u}\rangle\rvert$; optional C_{mix} (default: 0.1)
Stagnant thermal conductivity, k^0_{eff}	*Kunii*	φ_{K}, γ_{K}
(with *Wakao*)	*Krupiczka*, *ZS*	–
Wall-to-bed heat-transfer coefficient, h_w	*Const*	k_s, h_w
	Yagi, *Li*	k_s, $\lvert\langle\mathbf{u}\rangle\rvert$
	Bey	k_s, $\lvert\langle\mathbf{u}\rangle\rvert$, d_t

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